@henryleberre | **Henry LE BERRE /ɑ̃ʁiləbɛʁ/** | hberre3@gatech.edu

## EDUCATION

- **Georgia Institute of Technology** (Atlanta, GA, USA) | Bachelor's in Science in **Computer Science** — July 2021 – Present
  GPA: 3.84 | Concentration: **Modeling & Simulation** and **Systems & Architecture**
- **Lycée Lavoisier** (Paris Vè, France) — September 2018 – July 2021

## WORK EXPERIENCE

- **Space Exploration Technologies | Software Engineering Intern** — Los Angles, California, USA – Fall 2023
  Raptor Combustion Devices Engineering | Raptor Production Group

25% speed up of the Raptor Combustion CFD Code on production simulations through NVIDIA CUDA GPU kernel optimizations.

- **Verkada | Software Engineering Intern** — San Mateo, California, USA – Summer 2023
  Device Platform Team | Alarms and Intrusion
- **Computational Physics Group | Student Assistant** — Atlanta, Georgia, USA – Summer 2022

## PUBLICATIONS

- **[Preprint] Method for portable, scalable, and performant GPU-accelerated simulation of multiphase compressible flow** — May 2023
  Submitted to **arXiv** and **Computer Methods in Applied Mechanics and Engineering**. — 10.48550/arXiv.2305.09163
- **Early Application Experiences on a Modern GPU-Accelerated Arm-based HPC Platform** — September 2022
  HPC Asia '23. International Workshop on Arm-based HPC: Practice and Experience (IWAHPCE). — 10.1145/3581576.3581621

## TALKS, POSTERS, and PRESENTATIONS

- **Scalable GPU Accelerated Simulation of Multiphase Compressible Flow** — November 15, 2022
  *Supercomputing 2022 (SC'22), Dallas, Texas, USA* — Poster | Paper
- **Exascale for humans: A development environment for MFC** — February 19, 2022
  *Georgia Scientific Computing Symposium 2022 (GSCS 2022), Atlanta, Georgia, USA* — Website

## AWARDS

- **Verkadathon** (Verkada Hackathon) **Winner** ("Best Use of Verkada Products") — Summer 2023
- **President's Undergraduate Research Award** (**PURA**) **Travel** from the **Georgia Institute of Technology** — Fall 2022
- **Faculty Honors** and **Dean's List** from the **Georgia Institute of Technology**
- **National General Baccalaureate** (« Baccalauréat Général ») with **Highest Honors** (Mention « très bien ») — 2021
- **Highest Honors** (« Félicitations du conseil de classe ») for 9 consecutive trimesters — 2018 – 2021
- **Diplôme National du Brevet** with **Highest Honors** (Mention « très bien ») — 2018

## RESEARCH AFFILIATIONS

- **Computational Physics Group (CPG) at Georgia Tech | Multi-Component Flow Code (MFC)** — November 2021 – Present
- **Space Systems Design Lab (SSDL) at Georgia Tech | GT-II Satellite** — November 2021 – Dec. 2022

## LEADERSHIP & SERVICE

- **Student Volunteer** at **Supercomputing 2023 (SC'23)** | Denver, Colorado, USA — November 11 to 17 (2024)
- **Mentor** for team CEREAL at the **Georgia Tech Open Hackathon** (C++/MPI/Cuda/NVSHMEM) — January 2023
- **Lead** for **HPC & GPGPU**, **Halo-region compression**, and **Refactoring** for **MFC** — November 2022 – Present
- **Co-Chair** of the **Georgia Tech French Club** — August 2022 – Present

## MISCELANIOUS

- **Languages: French (Native | French Citizen), English (Native | U.S. Citizen), and German (High School Level)**